

DATABASE SYSTEMS

CHAP. 7: TRANSACTION PROCESSING CONCEPTS

Transaction

- Definition

- A transaction is a logical unit of program execution.

- Example

- Transfer of RwF100 between two accounts.

Typical OLTP Environments

- Airline/ railway reservation systems
- Banking systems (atm, eft, ...)
- Trading and brokerage systems
- Hotel / hospital systems
- Standard commercial systems

Recovery

- Hardware or software failures, result in an inconsistent database.
- The database must be restored.

Types of Failures

- Computer failure (system crash)
- Transaction or system error
- Local errors or exception conditions detected by the transaction
- Concurrency control enforcement
- Disk failure
- Physical problems and catastrophes

ACID Property of Transactions

- Atomicity
- Consistency preservation
- Isolation
- Durability or permanency

Atomicity

A transaction is an atomic unit of processing; it is either performed in its entirety or not performed at all.

Consistency preservation

- A transaction is consistency preserving if its complete execution take(s) the database from one consistent state to another.

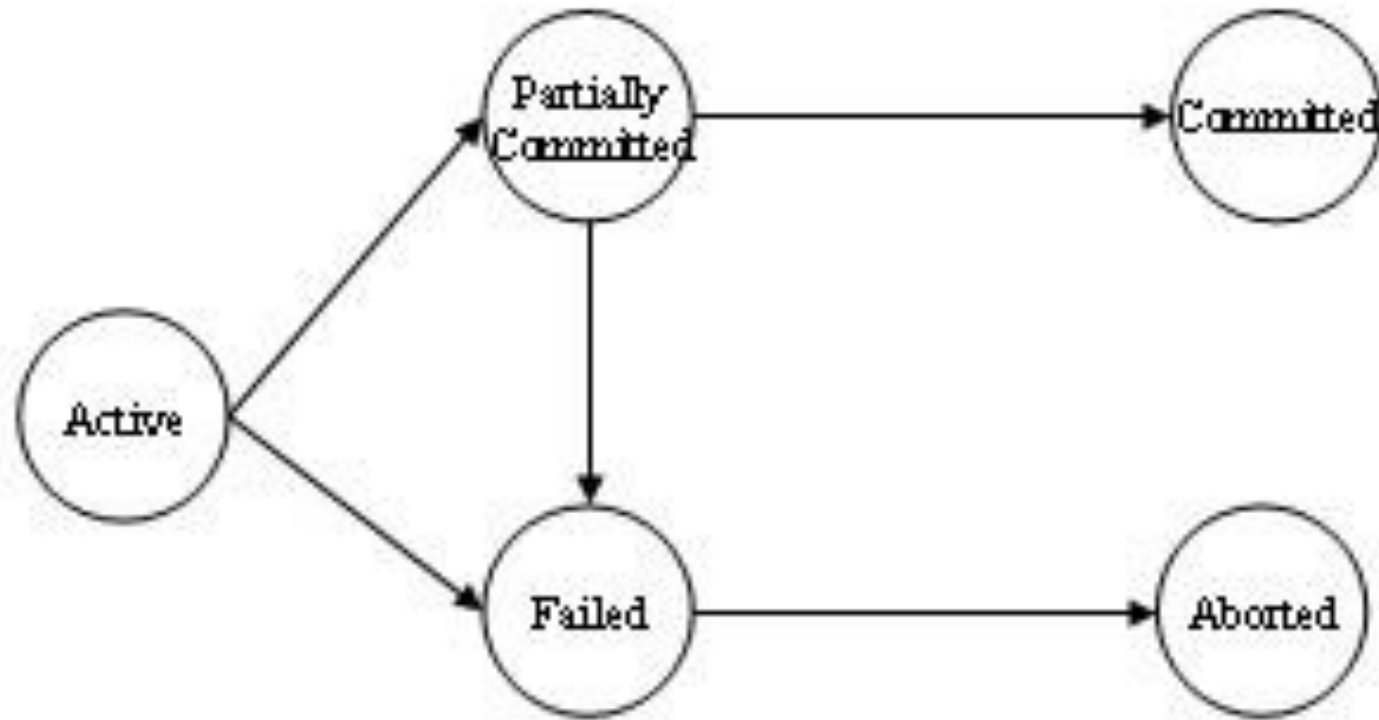
Isolation

- A transaction should appear as though it is being executed in isolation from other transactions.
- That is, the execution of a transaction should not be interfered by any other transactions executing concurrently.

Durability or permanency

- The changes applied to the database by a committed transaction must persist in the database.
- These changes must not be lost because of any failure.

Transaction States



Transaction States (Cont.)

- **Active**

- This is the initial state. The transaction stay in this state while it is executing.

- **Partially Committed**

- This is the state after the final statement of the transaction is executed.

- **Failed**

- This is the state after the discovery that normal execution can no longer proceed.

Transaction States (Cont.)

- **Aborted**

- The state after the transaction has been rolled back and the database has been restored to its state prior to the start of the transaction.

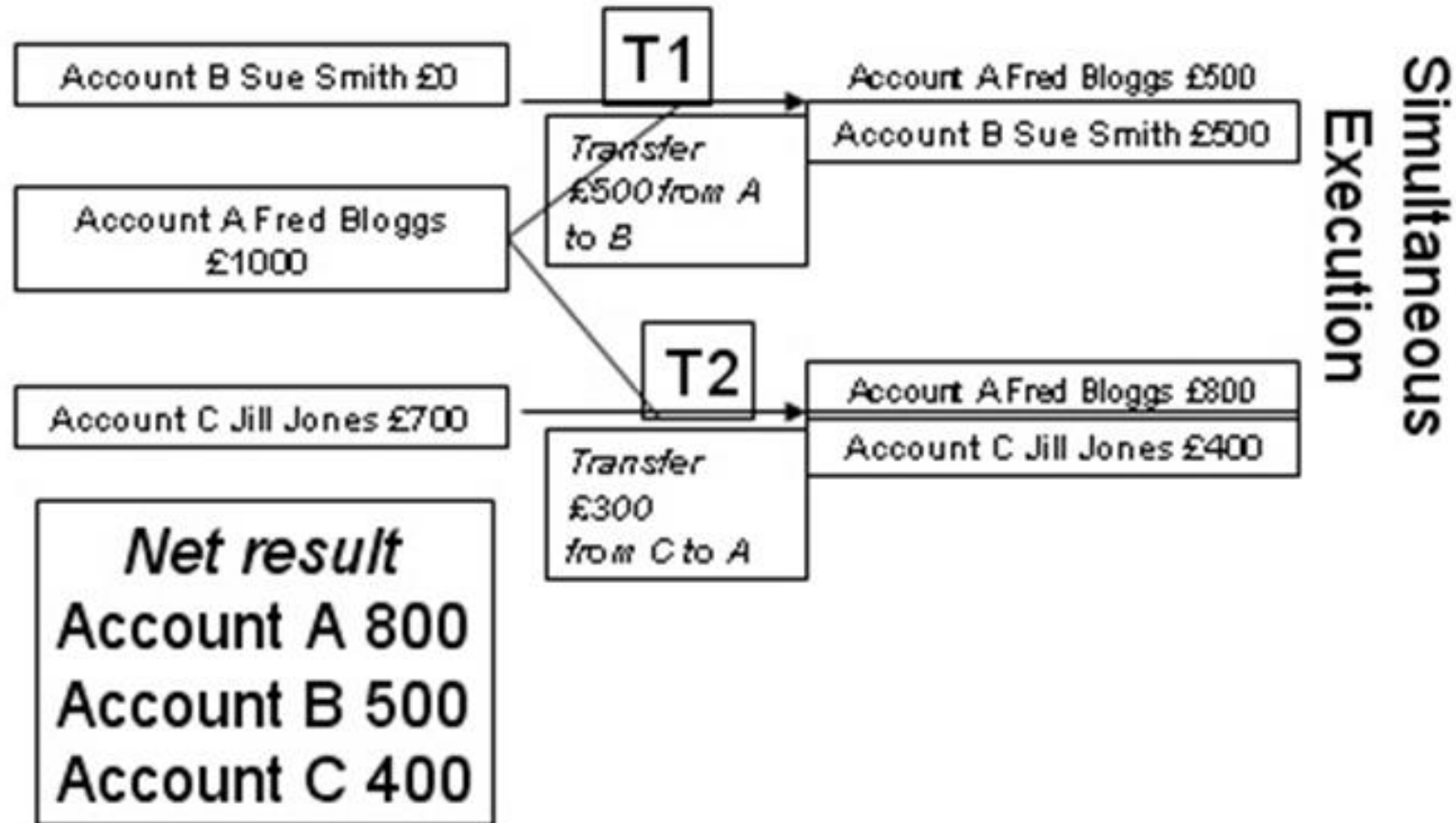
- **Committed**

- The state after successful completion of the transaction.

Transaction as a Concurrency Unit

- Transactions must be synchronised correctly to guarantee database consistency.

Transaction as a Concurrency Unit



Concurrent Executions

- Increased processor and disk utilization
- Reduced average response time

Concurrency control schemes

- Concurrency control schemes are designed to allow two or more transactions to be executed correctly while maintaining serial equivalence
 - Serial Equivalence is correctness criterion
 - Schedule produced by concurrency control scheme should be equivalent to a serial schedule in which transactions are executed one after the other

Concurrency control schemes(Cont.)

- Schemes:
 - Locking,
 - Optimistic concurrency control,
 - Time-stamp based concurrency control

Serializability

- Transactions are said to be serializable if the results of running transactions simultaneously are the same as the results of running them serially.

Conflicting Instructions

- Instructions l_i and l_j of transactions t_i and t_j respectively, **conflict** if and only if there exists some item q accessed by both l_i and l_j , and at least one of these instructions wrote q .

Conflicting Instructions (Cont.)

- $L_i = \text{read}(q)$, $l_j = \text{read}(q)$. L_i and l_j don't conflict
- $L_i = \text{read}(q)$, $l_j = \text{write}(q)$. They conflict
- $L_i = \text{write}(q)$, $l_j = \text{read}(q)$. They conflict
- $L_i = \text{write}(q)$, $l_j = \text{write}(q)$. They conflict

Conflict Serializability

- Let instructions i and j belonging to transactions t_1 and t_2 be executed consecutively by the DBMS.
- i and j can be swapped in their execution order if i and j refer to different data elements
- i and j can be swapped in their execution order **if** i and j refer to the same data element and both perform a **read** operation only.
- i and j are said to ***conflict*** if i and j belong to different transactions and at least one of them is a write operation.